

HDOC DAY-11 ASSIGNMENT

QUESTION

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Find the sum of squares of its digits using a while loop. Example: Number = 123, Sum = 14.
- Find the sum of cubes of its digits using a for loop. Example: Number = 123, Sum = 36.
- Find the difference between the sum of cubes and the sum of squares of its digits using a do-while loop. Example: Number = 123, Difference = 22.

Algorithm

1. Start.
2. Input a positive integer n.
3. Display the menu with three choices.
4. Input the user's choice.
5. Use switch-case to select the required operation.
6. For choice 1, set sum = 0 and repeatedly extract the last digit using a while loop. Add digit × digit to sum and remove the last digit from the number.
7. For choice 2, set sum = 0 and use a for loop to extract each digit. Add digit × digit × digit to sum.
8. For choice 3, calculate the sum of squares using a do-while loop and the sum of cubes using a second do-while loop. Find their difference as cubeSum – squareSum.
9. Display the result for the selected operation.
10. If the choice is invalid, display an appropriate message.
11. Stop.

C Program

```
#include <stdio.h>
int main()
{
    int n, choice, digit;
    int sumSquares, sumCubes, difference;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    printf("\nMENU\n");
    printf("1. Sum of squares of digits\n");
    printf("2. Sum of cubes of digits\n");
    printf("3. Difference between sum of cubes and sum of squares\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch(choice)
```

```

{
case 1:
sumSquares = 0;
while(n > 0)
{
digit = n % 10;
sumSquares = sumSquares + digit * digit;
n = n / 10;
}
printf("Sum of squares of digits = %d", sumSquares);
break;
case 2:
sumCubes = 0;
for(; n > 0; n = n / 10)
{
digit = n % 10;
sumCubes = sumCubes + digit * digit * digit;
}
printf("Sum of cubes of digits = %d", sumCubes);
break;
case 3:
sumSquares = 0;
sumCubes = 0;
do
{
digit = n % 10;
sumSquares = sumSquares + digit * digit;
sumCubes = sumCubes + digit * digit * digit;
n = n / 10;
}
while(n > 0);
difference = sumCubes - sumSquares;
printf("Difference between sum of cubes and sum of squares = %d", difference);
break;
default:
printf("Invalid choice!");
}
return 0;

```

Test Case Table

Test Case	Input Number	Choice	Operation	Expected Output	Actual Output	Status
1	123	1	Sum of squares	14	14	Pass
2	123	2	Sum of cubes	36	36	Pass

3	123	3	Cube sum – square sum	22	22	Pass
4	5	1	Sum of squares	25	25	Pass
5	45	2	Sum of cubes	189	189	Pass
6	10	3	Cube sum – square sum	0	0	Pass
7	123	4	Invalid choice	Invalid choice.	Invalid choice.	Pass

Compilation and Execution Evidence

```

kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ gedit question.c
** (gedit:6801): WARNING **: 13:38:03.544: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:6801): WARNING **: 13:38:03.544: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
(kali@kali)-[~/Desktop]
$ gcc question.c -o question
(kali@kali)-[~/Desktop]
$ ./question
Enter a positive integer: 123

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 1
Sum of squares of digits = 14

(kali@kali)-[~/Desktop]
$ ./question
Enter a positive integer: 123

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 2
Sum of cubes of digits = 36

(kali@kali)-[~/Desktop]
$ ./question
Enter a positive integer: 123

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 3

```

```

kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ ./question
Enter a positive integer: 123

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 3
Difference between sum of cubes and sum of squares = 22

(kali@kali)-[~/Desktop]
$ ./question
Enter a positive integer: 5

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 1
Sum of squares of digits = 25

(kali@kali)-[~/Desktop]
$ ./question
Enter a positive integer: 45

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 2
Sum of cubes of digits = 189

(kali@kali)-[~/Desktop]
$ ./question
Enter a positive integer: 10

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares

```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ ./question
Enter a positive integer: 10

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 3
Difference between sum of cubes and sum of squares = 0

(kali@kali)-[~/Desktop]
$ ./question
Enter a positive integer: 123

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference between sum of cubes and sum of squares
Enter your choice: 4
Invalid choice!

(kali@kali)-[~/Desktop]
$
```